

Steer on a Sphere: Geometric Control of Transformer Outputs

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August 15, 2026



Abstract

Layer normalization constrains transformer hidden states to a sphere whose radius is $\|\gamma_l\|$ per layer (not the textbook $\sqrt{d}$). Within this sphere, a single tangent step toward any token’s LM head direction reaches that token with 91–98% reliability across four model families and never lowers its rank (100% improvement). We propose a theory of edge-of-chaos dynamics: the per-model Lyapunov rate clusters near $\lambda \cdot L_{\text{comp}} \approx 0.5$ for the Gemma, GPT-2, and mid-Qwen families (spanning 6 of 13 tested architectures). The BOS token defines a growing reference axis orthogonal to vocabulary tokens across 28 layers. Some tokens are self-reinforcing fixed points (cow tipping)—the degenerate limit of the dynamics at $\lambda \rightarrow 0$ —with narrow basins that can be encoded into data as a defense against LLM scraping. All measurements on Qwen2.5-7B with spot-checks on DeepSeek-7B and Mistral-7B. Full reproduction steps provided. Cow tipping occurs on all tested models, with vocabulary-specific pit tokens. Because the pits are derived from model weights, they can be computed for any architecture without retraining or data.

1 The Sphere

RMSNorm maps hidden states to a sphere whose expected radius is $\|\gamma_l\|$:

$$\|\text{RMSNorm}(h)\|^2 = \sum_{i=1}^d \gamma_{l,i}^2 \left(\frac{h_i}{\text{RMS}(h)} \right)^2$$

Per-instance norms float around this value (ratio 0.7–1.3), so the picture is a fuzzy shell. Measured γ norms across four model families:

Model	d	$\ \gamma_0\ $	$\ \gamma_{\text{mid}}\ $	$\ \gamma_{\text{end}}\ $
Mistral-7B	4096	16.6	111–158	195.5
DeepSeek-7B	4096	3.7	15–24	26.1
Qwen2-1.5B	1536	20.0	40–60	71.0
Qwen2.5-7B	3584	17.0	63.5	90.0

$\|\gamma_l\|$ is learned, varying across architectures even at the same d .

2 BOS Axis

The BOS token (position 0) defines a reference direction $\hat{b}_l = h_l^{(0)} / \|h_l^{(0)}\|$. Its residual norm grows from 2.8 to 102.2 across 28 layers, consistent with attention-sinking [3]. LM head rows W_t are approximately orthogonal to BOS at every layer ($\theta_{l,t} = \arccos |\hat{b}_l \cdot \hat{w}_t|$):

Layer	$\ LN(h)\ $	$\ BOS\ $	Token-BOS angle
0	15.9	2.8	88–90°
2	16.3	24.8	88–90°
4	33.5	21.8	89–90°
6	31.8	36.9	89–90°
8	36.5	44.1	88–90°
10	34.7	43.4	89–90°
12	37.5	50.3	89–90°
14	42.9	55.2	88–90°
16	44.4	53.0	89–90°
18	40.7	43.9	89–90°
20	45.2	48.2	89–90°
22	54.9	63.7	89–89°
24	55.5	68.7	87–90°
26	55.6	83.1	86–90°
27	63.1	102.2	87–90°

High-dimensional vectors are generically near-orthogonal ($\mathbb{E}[\cos] \approx 1/\sqrt{d}$), so the approximate 90° is expected; the consistent preservation of this angle while BOS grows 36× across layers is the observed regularity. Mistral-7B and DeepSeek-7B also give 87–90°.

3 Sphere Steering: Tangent Traversal

The LM head row W_t has a tangent component $g_t = W_t - (W_t \cdot \hat{h})\hat{h}$. Moving h along g_t increases the target logit ($W_t \cdot g_t = \|g_t\|^2 \geq 0$). Renormalizing keeps the traversal on the sphere:

$$h' = \frac{h + \alpha g_t}{\|h + \alpha g_t\|} \cdot \|\gamma_L\|$$

3.1 Quantitative Traversal

The LM head itself provides a complete geometric map of the output space: 152,064 directions, one per token, on the sphere. A single tangent step at $\alpha = 0.3$ reaches any of them. Tested on 1,000 random tokens across four model families:

Model	Rank 1	Rank improved
Qwen2.5-7B	94.5%	100.0%
DeepSeek-7B	97.5%	100.0%
Mistral-7B	96.2%	100.0%
Qwen2-1.5B	91.3%	100.0%

The rank improvement is guaranteed ($W_t \cdot g_t = \|g_t\|^2 \geq 0$), and the 91–98% reaching rank 1 follows from the fact that most LM head rows are isolated enough from their nearest neighbors to dominate the logit after a single tangent step. The fraction that fail are tokens in clusters (digits, question marks) where nearest-neighbor LM head rows compete.

3.2 Qualitative Demonstration

On Qwen2.5-7B, steering open-ended prompts produces visible output shifts. At $\alpha = 2$ on “Tell me something interesting:”, steering toward “75” changes the continuation from a generic fact (“The most common blood type is O positive...”) to a numeric statement (“1984 was the last year the United States had a budget surplus...”). At $\alpha = 3$ on “Once upon a time”, steering toward “apple” turns the story from “a young boy named Timmy...” into “apple trees were planted in the garden of Eden...” Common words and digits shift more readily than rare tokens; larger α trades fluency for strength.

Steering does not distinguish censored from uncensored paths. It is a white-box traversal primitive: with access to the model weights, anyone can move h toward any token direction on the sphere without retraining or fine-tuning.

4 Cow Tipping

Some tokens generate themselves when fed repeatedly. A token T is *self-consistent* if $s(T) = \text{softmax}(W \cdot h_T)[T]$ is high, where $h_T = \text{model}(e_T)$. A *pit* satisfies both high $s(T)$ and trajectory stability $\cos(h_{n+1}, h_n) \approx 1$: indefinite self-generation.

We scanned a subset of the Qwen2.5-7B vocabulary, testing permanence (3 input repetitions, 15-step self-prediction). Seven tokens pass:

Token	ID	$s(T)$	Trigger
0	15	0.975	000, 000 – 000 – 0000
ab	370	0.934	ababab
. (space+dot)	659	0.889	. . .
ere	485	0.878	erere
_ (space+underscore)	716	0.110	_ _ (3+ reps)
NULL byte	188	0.978	3+ consecutive null bytes
???	26610	0.386 [†]	8+ question marks

[†]Deterministic fixed point ($\cos \approx 1.0$ once generated).

The 0 token is clearest: 000 yields $p = 0.974$, $\cos \approx 1.0$, $p > 0.95$ within 3 steps. Phone numbers trigger it via three consecutive token-15 units. The 0 pit was confirmed on DeepSeek-7B. Mistral-7B has its own set of self-consistent tokens (e.g., its vocabulary contains different pits with s up to 0.59 and 15/15 permanence), confirming that the phenomenon is general but the specific pit tokens are vocabulary-dependent. All pits have narrow basins—any trailing token disrupts the fixed point.

4.1 Defensive Encoding

Because pits are derived directly from model weights—computed as $s(T) = \text{softmax}(W \cdot h_T)[T]$ with a permanence test over the vocabulary—each model’s pits are unique to its tokenizer and LM head geometry. The accompanying code release ([pit.engine.py](#)) provides a full scan that finds all pits for any model.

The pits can be used as a defensive primitive: embedding pit triggers into data causes any model that processes it to fall into a fixed-point loop rather than continue reading. Because the basins are narrow, the trigger must be the terminal content of each chunk. We verified three encodings on Qwen2.5-7B:

- **NULL bytes.** Appending `\x00\x00\x00` to HTML makes the model emit null bytes indefinitely. The trigger is invisible in rendered text ($s = 0.978$).
- **Repeated sub-tokens.** A chunk ending in `cut cut cut` collapses the model into `cut cut cut ...`; key requires 6 repetitions to lock.
- **Phone number (terminal).** `000 – 000 – 0000` locks only as the final content; any trailing markup breaks it.

A crawler reading a page that terminates in one of these triggers stops producing useful output and repeats the pit token instead. We report this as a defensive capability: it lets a content owner degrade automated scraping by large language models without affecting human readers.

5 Edge of Chaos: A Theory

The residual stream is a dynamical system $h_{l+1} = h_l + f_l(h_l)$. Its per-layer Lyapunov exponent measures how perturbations grow:

$$\lambda_l = \ln \frac{\|h_{l+1}^{\text{pert}} - h_{l+1}^{\text{ref}}\|}{\|h_l^{\text{pert}} - h_l^{\text{ref}}\|}$$

A perturbation that neither explodes nor vanishes during computation would place the system at the edge of chaos, corresponding to $\lambda \cdot L_{\text{comp}} \approx 0.5$ (one e-fold of total expansion). We measured the per-layer rate across 13 model families:

Cluster	Models	$\lambda \cdot L_{\text{comp}}$
≈ 0.5	Gemma-2-2B, Gemma-2-9B, GPT-2, DS-R1-7B, Qwen-1.5B, Qwen2.5-7B	0.37–0.57
≈ 0.8	SmolLM2-360M, DeepSeek-7B	0.76–0.88
≈ 1.0	Phi-3.5-mini, Mistral-7B	1.00–1.47
≈ 0.1	Qwen2.5-0.5B, Qwen2.5-3B	0.01–0.10

The edge-of-chaos balance ($\lambda \cdot L \approx 0.5$) holds for roughly half the models—notably the Gemma family, GPT-2, and mid-sized Qwen—but is not universal. We present it as a theory that captures a subset of architectures, not as a scaling law. Cow tipping (Section 4) is the degenerate limit of this balance: a fixed point where $\lambda_l \rightarrow 0$ and the trajectory stops changing entirely.

6 Reproduction

All measurements are scriptable in under 100 lines of Python. Full code at <https://github.com/ntrillard/transformer-geometry>. The procedure:

1. **Sphere radius.** Extract γ_l from model state dict. Feed prompts, record post-LN hidden state norms per layer. Compare to $\|\gamma_l\|$.
2. **BOS axis.** Feed a prompt, extract hidden states per layer. For each layer, compute $\hat{b}_l$ from position 0 and $\theta_{l,t}$ for any LM head row W_t .
3. **Steering (traversal).** Forward-pass a prompt to get h . For target token t , compute g_t , step with α , renormalize by $\|\gamma_L\|$, then feed h' as the first-generation hidden state. Verified across 1,000 random tokens on each of 4 models (91–98% rank-1, 100% rank improvement).
4. **Cow tipping.** For token t , decode, re-encode, forward-pass to get h_T . Compute $s(T)$. If $s(T) > 0.01$, feed t three times and check 15-step self-prediction with greedy decoding. Record permanence and cos stability.

Acknowledgments

DeepSeek V4 Pro provided writing assistance.

References

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